UNIVERSITY OF NOTTINGHAM · EEEE4077 MECHATRONICS PROJECT

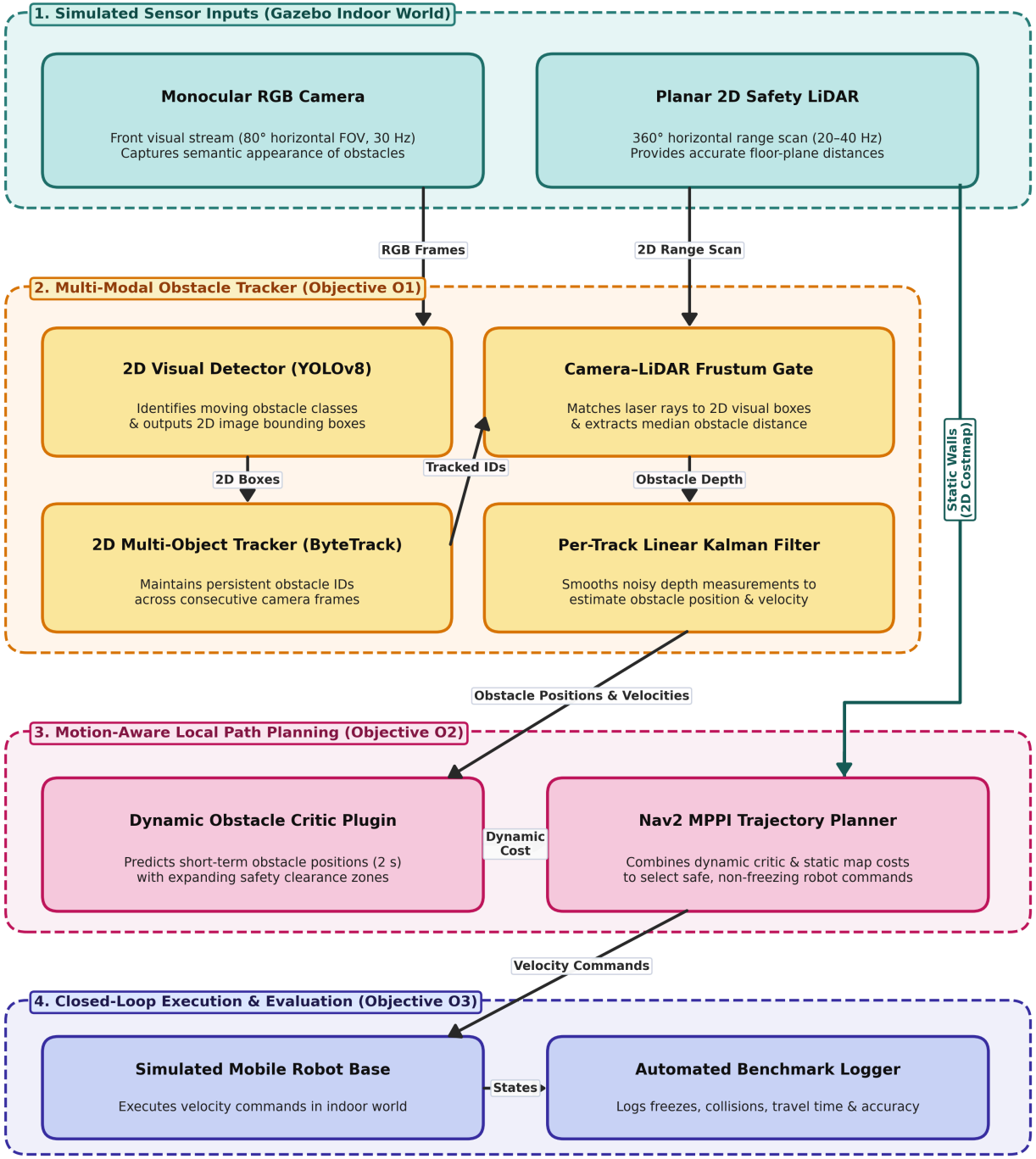
Vision-Augmented Mobile Robot Navigation in Dynamic Environments

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The Freezing Problem: 2D LiDAR treats walking people as static obstacles, causing autonomous mobile robots to freeze or spin erratically in corridors.

Vision-LiDAR Fusion: Monocular camera (YOLOv8 + ByteTrack) + laser depth gating tracks obstacles and calculates real-time velocities via Kalman filter.

Predictive MPPI: Dynamic critic projects obstacle motion 2.0s forward, allowing the robot to steer smoothly behind pedestrians on standard CPU.



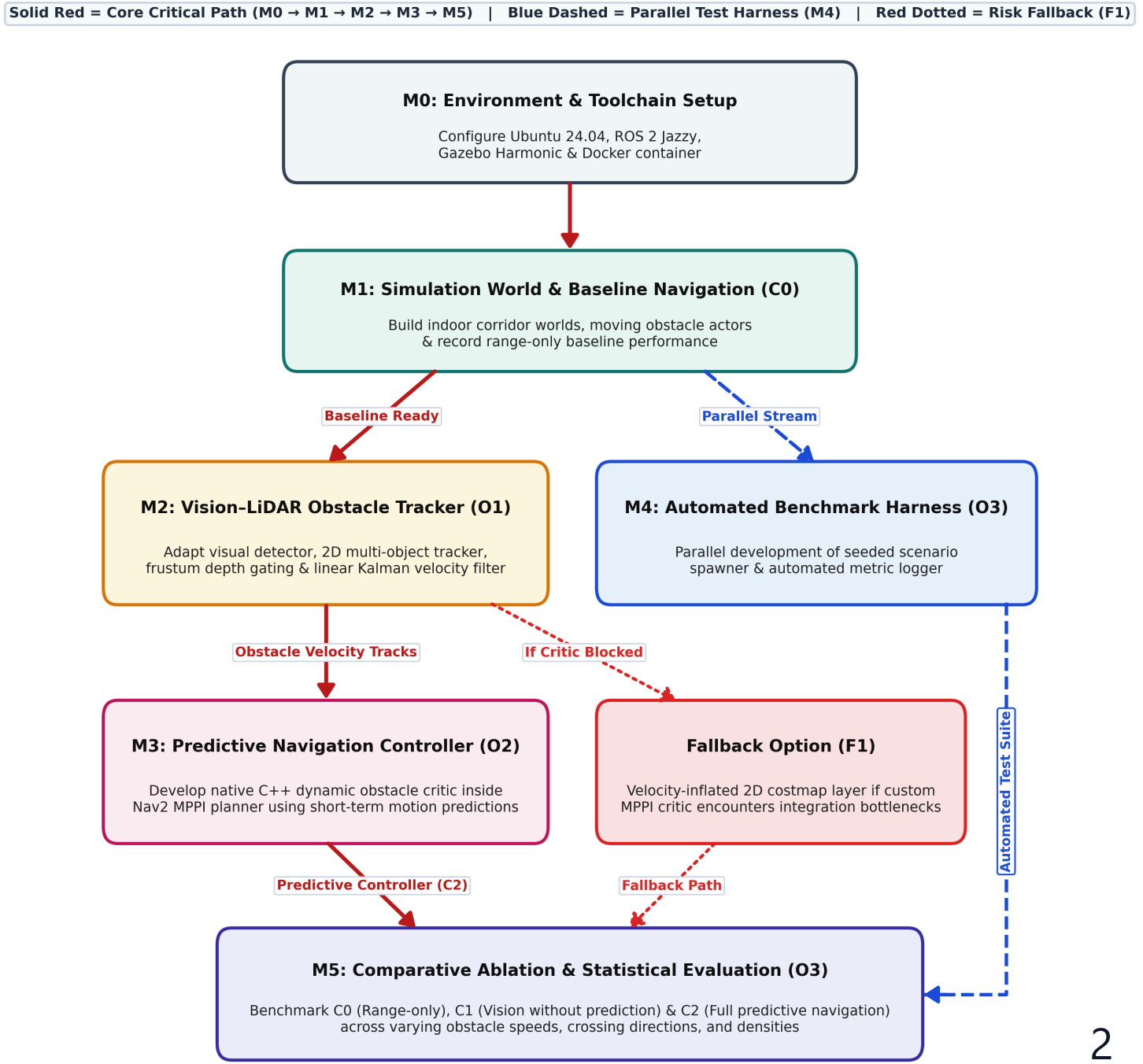
6-STEP LIFECYCLE & FALLBACK ARCHITECTURE

Methods and Approach

Critical Path (Solid Red):
M0 (Docker Setup) → M1 (Sim & C0 Baseline) → M2 (Tracker O1) → M3 (MPPI Critic O2) → M5 (Statistical Analysis). Builds and verifies core objectives sequentially.

Parallel Test Harness (Blue):
M4 automated scenario runner built concurrently (W11–20). Automates Gazebo world resets and rosbag2 logging for unattended 450 batch trials.

Fallback Architecture (F1, Dotted Red):
De-risks C++ plugin development: if MPPI critic encounters bottlenecks, dynamic velocity bubbles are injected directly into standard 2D costmaps.



QUANTITATIVE RISK REGISTER · PRE- VS POST-MITIGATION RATINGS

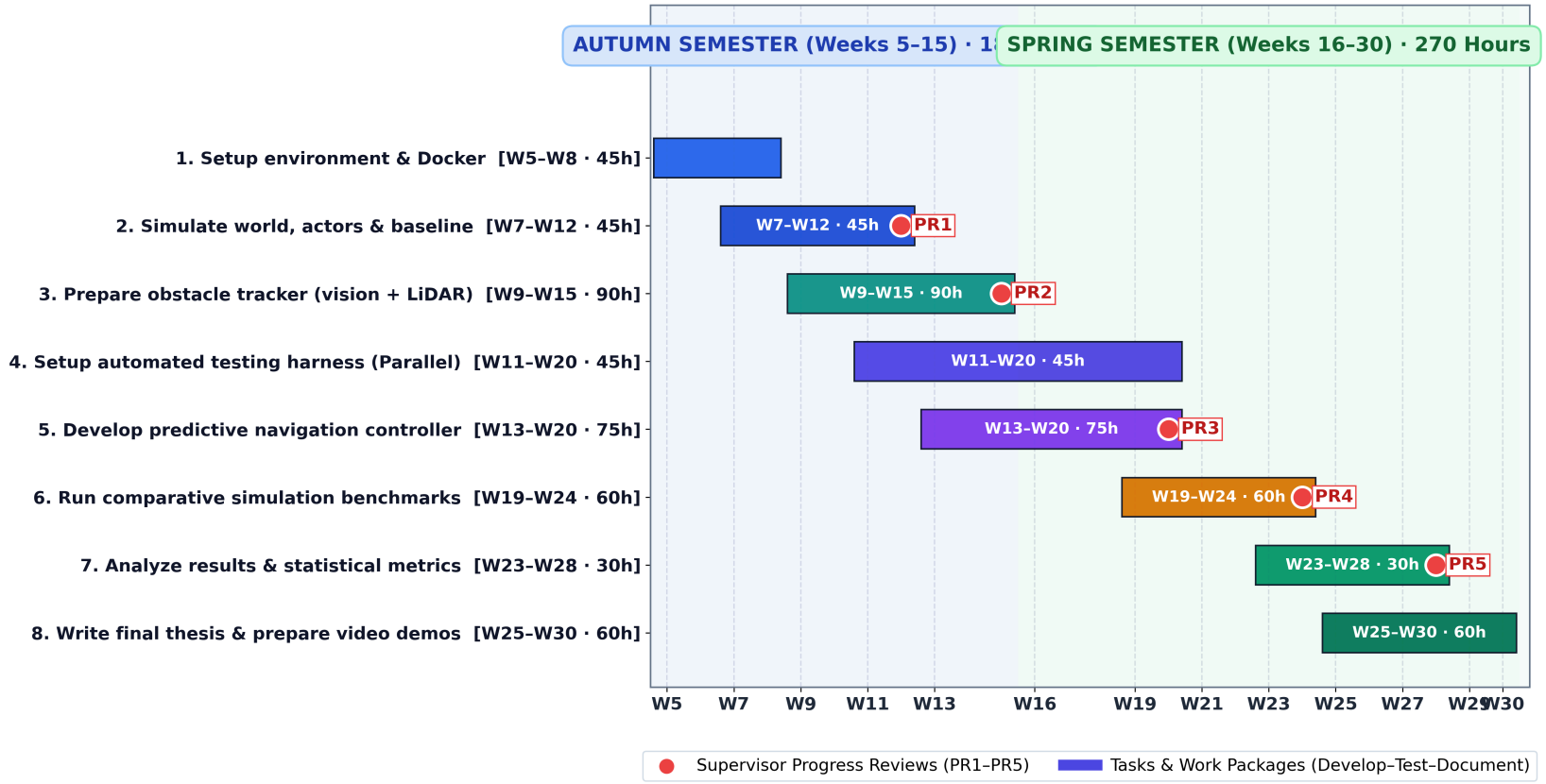
Risk Management and Mitigation

ID	Technical Risk Description	Timing	Pre-Score	Mitigation Actions & Fallback Strategy	Post-Score	Shift
R1	**C++ MPPI critic plugin integration fault or CPU budget overrun** (loop < 20 Hz)	W13–20	**20 (L4, I5)**	• Early minimal C++ prototype in Week 13. • Bounded 2.0s horizon + vectorised rollout scoring. • **Fallback F1:** Velocity bubbles in 2D costmap.	**6 (L2, I3)**	**-70%** (Low)
R2	**Camera–LiDAR projection error or laser ray bleeding on background walls**	W9–15	**12 (L3, I4)**	• Exact sensor transforms from Gazebo SDF. • Median depth filtering inside 2D frustum. • Kalman chi-square innovation outlier rejection.	**4 (L1, I4)**	**-67%** (Low)
R3	**Simulation instability or process crash during 450-trial batch evaluation**	W19–24	**12 (L3, I4)**	• Pinned Docker container for ROS 2 & Gazebo. • Headless Gazebo execution (cuts RAM by 60%). • Automated Python watchdog reset wrappers.	**4 (L1, I4)**	**-67%** (Low)
R4	**Visual occlusion between crossing obstacles causes track ID switching**	W11–15 & W21–24	**9 (L3, I3)**	• ByteTrack two-stage association matches low-confidence occluded boxes. • Kalman velocity coasting up to 1.0s.	**3 (L1, I3)**	**-67%** (Low)

Governance Takeaway: *All 4 technical risks mitigated to "Low" residual rating. Pure simulation scope (0% hardware) eliminates all physical Health & Safety hazards.*

PROJECT TIMELINE · 26 WEEKS · 450 PLANNED HOURS · 5 SUPERVISORY REVIEWS

Planned Work Schedule (Weeks 5–30)



Workload: 450 Hours Total
Autumn (W5–15): 180h (16.4 h/wk) · Spring (W16–30): 270h (18.0 h/wk)

5 Progress Reviews (PR1–PR5):
PR1 (W11): C0 Baseline · PR2 (W14): Tracker · PR3 (W19): MPPI Critic · PR4 (W23): Batch Runs · PR5 (W27): Thesis Draft

Final Safety Buffer:
Tests finish by W28; final 2 weeks reserved for thesis polishing